In and out: surface area, diffusion and osmosis

9C Threat from disease · Lesson 7 of 7

Date _____ Class _____

Today I can 今天我能

- ☐ I can calculate a surface area : volume ratio and explain why it matters for cells and organisms.
- ☐ I can describe how osmosis happens through a partially permeable membrane and use it to explain what I see.

What happens today

1. Do Now: a 1 cm cube
2. Learn: Surface area : volume ratio · Why the ratio matters · and more
3. Practical: Osmosis in potato strips
4. Challenge, then the exit ticket

Student Book pages 50 to 52 · Workbook pages 36 to 38

What I learned today (one or two sentences) 今天我学到了：

Key Vocabulary

Term	中文	Definition
surface area	表面积	the total area of the outside of an object
volume	体积	the amount of space an object takes up
ratio	比	one quantity compared with another, e.g. 6 : 1
diffusion	扩散	the overall movement of particles from where they are more concentrated to where they are less concentrated
concentration gradient	浓度梯度	a difference in concentration between two places
osmosis	渗透	the diffusion of water molecules through a partially permeable membrane
partially permeable membrane	半透膜	a membrane with tiny holes that let small molecules through but not large ones
cell membrane	细胞膜	the partially permeable layer around a cell
unicellular	单细胞的	made of one cell
villi	绒毛	finger-like folds that give the small intestine a large surface area
pandemic	大流行病	an outbreak of a disease across the whole world

Command words 指令词: Calculate 计算 · Complete 完成（填写） · Explain 解释 · Give 给出 · State 说明（写出） · Suggest 提出（建议） · Tick 打勾

Quick Review

a 1 cm cube

A cube has sides of 1 cm.

1 Calculate its surface area. (It has 6 faces.)

2 Calculate its volume.

3 Which gas diffuses into cells so they can respire?

Core 1 of 2

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 Complete the text using words from the word bank.

The movement of all particles is _____. Where there is a concentration _____, more particles move out from a region of _____ concentration than from a region of _____ concentration. This spreading out is known as _____.

2 Complete the table for three cubes.

Cube	Surface area (cm2)	Volume (cm3)	Surface area : volume ratio
sides 1 cm	6	1	6 : 1
sides 2 cm		8	
sides 3 cm	54		

Core 2 of 2

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

3 Potato strips in water and salt solution.

a) Record your results.

Strip	Liquid	Length at start (mm)	Length at end (mm)	Change (mm)	Firm or floppy?
A	pure water				
B	salt solution				

b) Tick the strip that gained water.

- ☐ A strip A (water)
- ☐ B strip B (salt solution)

c) Explain your results using the word osmosis.

Challenge 1 of 3

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 A silky anteater has a mass of about 400 g and eats up to 5000 ants a day. A giant anteater has a mass of about 40 kg and eats up to 30 000 ants a day.

a) Calculate the number of ants eaten per gram of body mass for each anteater.

b) Suggest a reason for the difference, using ideas about surface area : volume ratio.

Challenge 2 of 3

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C2 Osmosis in living things.

a) Paramecium is a unicellular organism that lives in fresh water. It has a contractile vacuole that collects water from the cytoplasm and pumps it out. Explain why Paramecium needs to remove water from its cytoplasm.

Challenge 3 of 3

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C2 *continued*

b) Tropical sprue stops food being digested properly, so undigested food stays dissolved inside the small intestine. Explain how this can cause diarrhoea (watery faeces) by osmosis.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is osmosis?

2 A 2 cm cube has a surface area of 24 cm2 and a volume of 8 cm3. State its surface area : volume ratio.

3 Give one reason why rapid international travel makes a pandemic more likely.

Today I can: Goal 1 ☐ Goal 2 ☐